PVsyst - Simulation report

Grid-Connected System

Project: 100 MW Solar PV Grid Connected

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 108.7 MWp

Dholera - India

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PVsyst V8.1.3
VC0, Simulation date:
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with V8.1.3

Project summary

Geographical Site		**Situation**		**Project settings**	
Dholera		Latitude	22.2493 °(N)	Albedo	0.20
India		Longitude	72.1942 °(E)		
		Altitude	14 m		
		Time zone	UTC+5.5		

Weather data
Dholera
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System
Simulation for year no 10

No 3D scene defined, no shadings

Orientation #1		**Near Shadings**	**User's needs**
Fixed plane		no Shadings	Unlimited load (grid)
Tilt/Azimuth	21 / 0 °		

System information

PV Array		**Inverters**	
Nb. of modules	190689 units	Nb. of units	50 units
Pnom total	108.7 MWp	Total power	100000 kWac
		Pnom ratio	1.09

Results summary

Simulation step	Hourly				
Produced Energy	169607279 kWh/year	Specific production	1560 kWh/kWp/year	Perf. Ratio PR	77.44 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step
Hourly

Orientation #1
Fixed plane
Tilt/Azimuth 21 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX
(Custom parameters definition)	
ReNew_M10_144_570_TOPCon_Bifacial_RPS2MH72BDXXX.PAN	
Unit Nom. Power	570 Wp
Number of PV modules	190689 units
Nominal (STC)	108.7 MWp
Modules	11217 units x 17 In series
At operating cond. (50°C)	
Pmpp	99.99 MWp
U mpp	688 V
I mpp	145362 A

Inverter

Manufacturer	Generic
Model	Solar Inverter DelCEN 1000x2
(Original PVsyst database)	
Unit Nom. Power	2000 kWac
Number of inverters	50 units
Total power	100000 kWac
Operating voltage	610-930 V
Max. power (=>40°C)	2200 kWac
Pnom ratio (DC:AC)	1.09

Total PV power

Nominal (STC)	108693 kWp
Total	190689 modules
Module area	492598 m²
Cell area	454780 m²

Total inverter power

Total power	100000 kWac
Max. power	110000 kWac
Number of inverters	50 units
Pnom ratio	1.09

Array losses

Array Soiling Losses
Loss Fraction 2.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 0.078 mΩ
Loss Fraction 1.50 % at STC

LID - Light Induced Degradation
Loss Fraction 2.00 %

Module Quality Loss
Loss Fraction -0.22 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.15 %

Module average degradation
Year no 10
Loss factor 0.4 %/year
Imp / Vmp contributions 80% / 20%
Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

Array losses

Spectral correction
FirstSolar model
Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0.85914	-0.02088	-0.0058853	0.12029	0.026814	-0.001781

System losses

Unavailability of the system
Time fraction 1.0 %
3.7 days,
3 periods

Auxiliary losses

AC wiring losses

Inv. output line up to injection point
Inverter voltage 400 Vac tri
Loss Fraction 0.00 % at STC

Inverter: Solar Inverter DelCEN 1000x2

Wire section (50 Inv.) Copper 50 x 3 x 4000 mm²
Average wires length 0 m

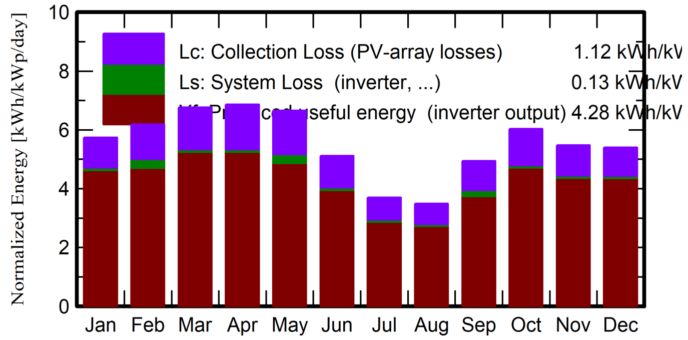
Main results

Simulation step
Hourly

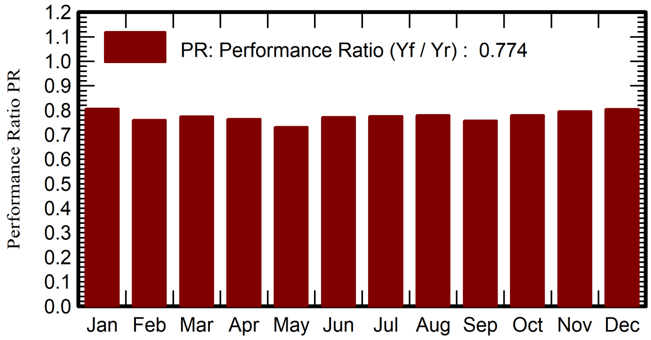
System Production

Produced Energy	169607279 kWh/year	Specific production	1560 kWh/kWp/year
		Perf. Ratio PR	77.44 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	138.6	41.00	21.43	177.7	171.2	15854950	15565914	0.806
February	145.4	49.86	24.34	173.2	166.8	15230491	14301346	0.760
March	191.6	63.86	28.71	209.7	201.8	17970038	17657618	0.775
April	204.6	75.30	32.33	205.9	197.6	17402053	17095003	0.764
May	218.8	87.88	33.65	206.1	197.6	17397829	16378943	0.731
June	165.5	98.20	32.10	153.4	146.6	13143288	12878977	0.773
July	122.2	81.18	29.54	114.6	109.3	9916000	9672160	0.776
August	111.9	78.39	28.50	108.1	103.1	9402126	9166708	0.780
September	142.9	82.05	29.27	148.1	142.1	12875487	12192304	0.757
October	163.3	63.25	30.09	186.9	179.6	16134777	15851384	0.780
November	132.7	55.29	26.74	164.3	158.0	14461248	14204406	0.796
December	128.0	41.60	22.80	167.3	161.2	14915898	14642516	0.805
Year	1865.6	817.86	28.30	2015.1	1934.9	174704186	169607279	0.774

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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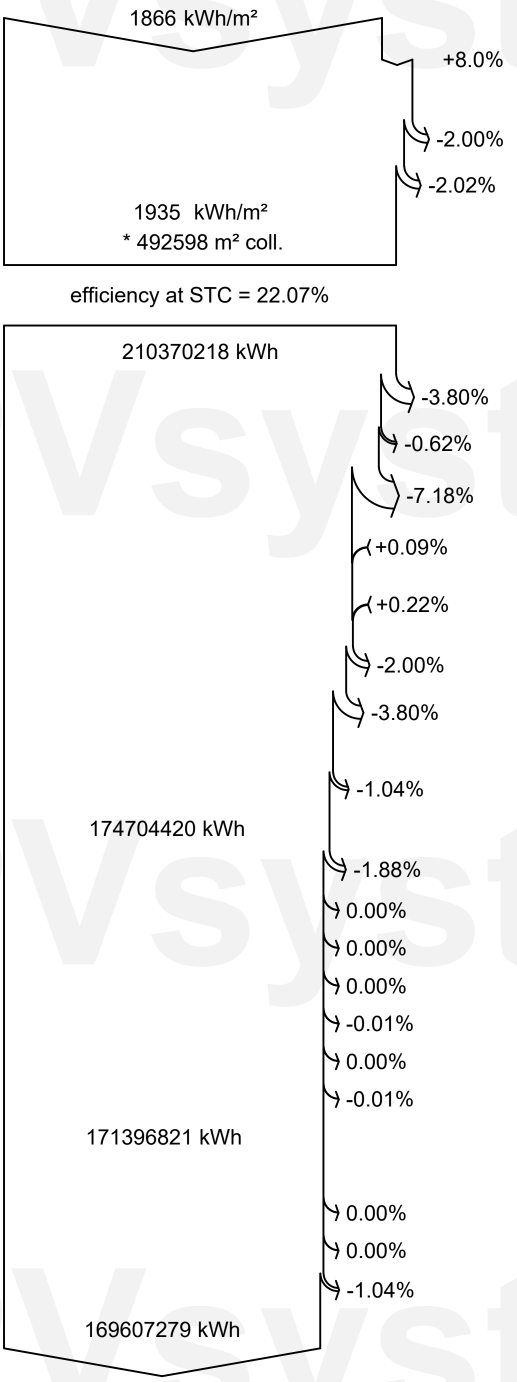
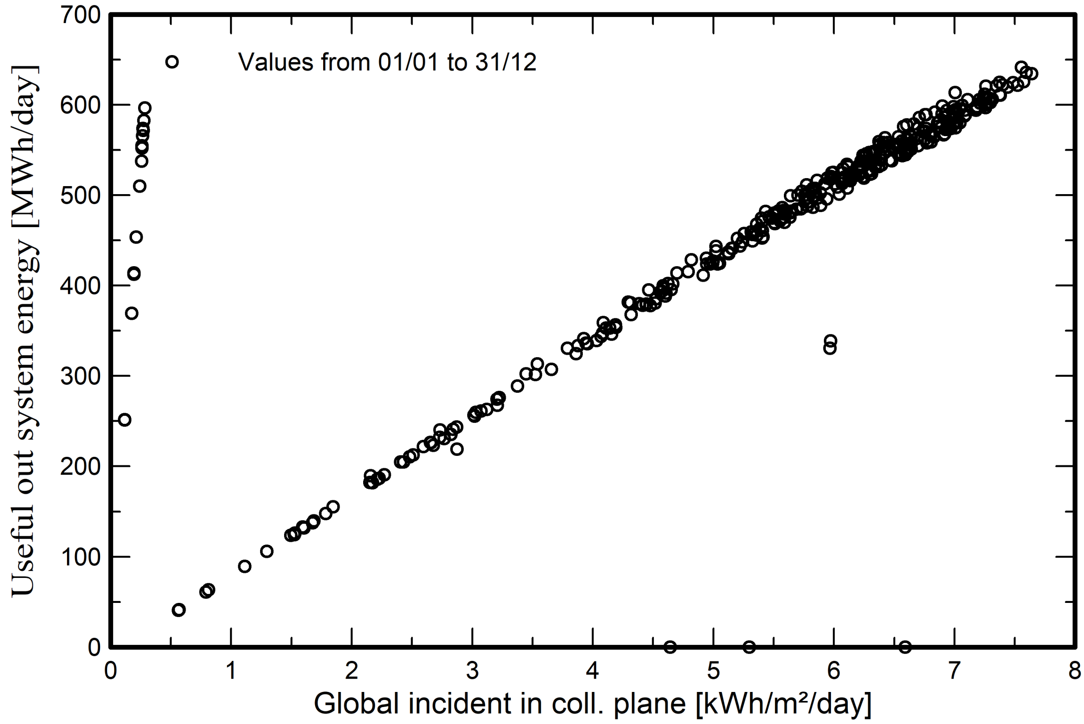
Loss diagram



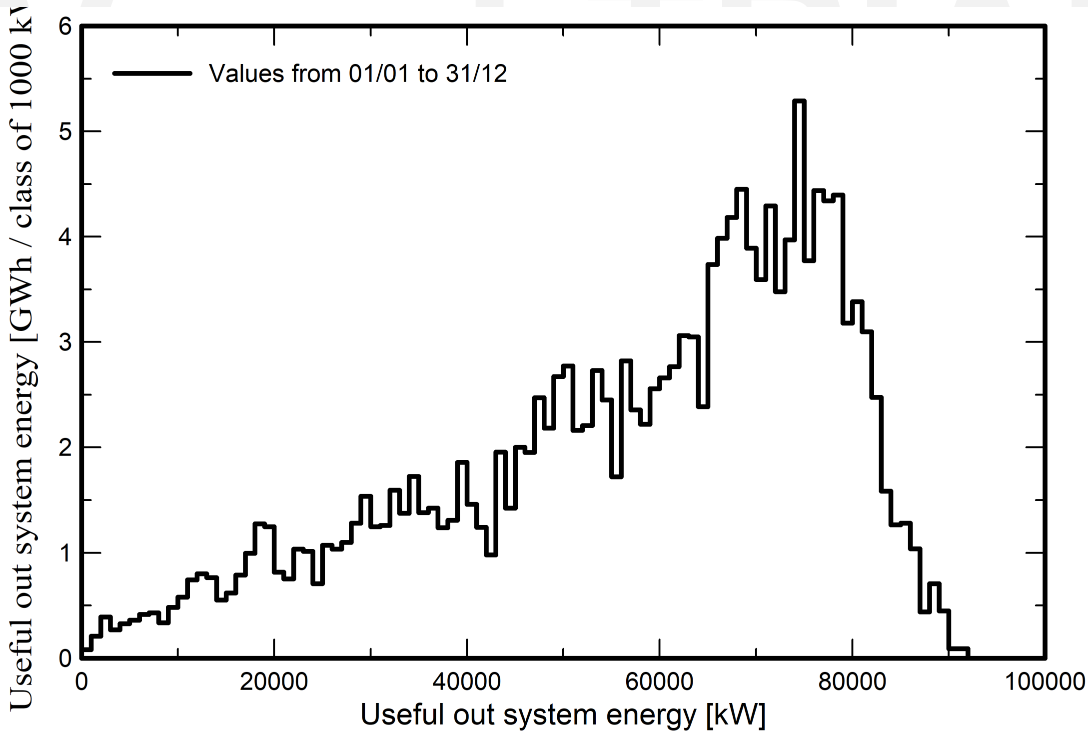
Global horizontal irradiation
Global incident in coll. plane
Soiling loss factor
IAM factor on global
Effective irradiation on collectors
PV conversion
Array nominal energy (at STC effic.)
Module Degradation Loss (for year #10)
PV loss due to irradiance level
PV loss due to temperature
Spectral correction
Module quality loss
LID - Light induced degradation
Mismatch loss, modules and strings
(including 1.7% for degradation dispersion)
Ohmic wiring loss
Array virtual energy at MPP
Inverter Loss during operation (efficiency)
Inverter Loss over nominal inv. power
Inverter Loss due to max. input current
Inverter Loss over nominal inv. voltage
Inverter Loss due to power threshold
Inverter Loss due to voltage threshold
Night consumption
Available Energy at Inverter Output
Auxiliaries (fans, other)
AC ohmic loss
System unavailability
Energy injected into grid

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Single-line diagram

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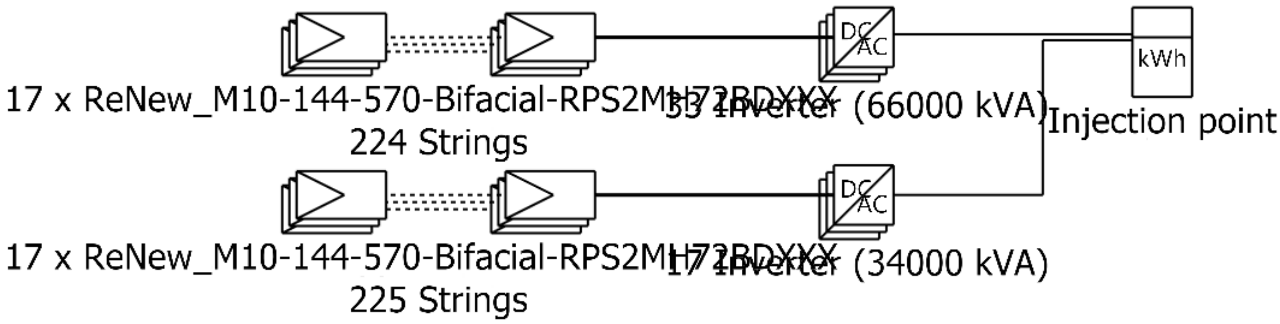

PV module	ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX
Inverter	Solar Inverter DelCEN 1000x2
String	17 x ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX

100 MW Solar PV Grid Co
nnected

VC0 : New simulation variant

31/05/26